

WPI



SCAPEgoat Side-channel Analysis Library

Dev Mehta, Trey Marcantino, Mohammad Hashemi, Sam Karkache, Dillibabu Shanmugam,
Patrick Schaumont, Fatemeh Ganji

Accepted and presented at the 2025 IEEE 43rd VLSI Test Symposium (VTS)

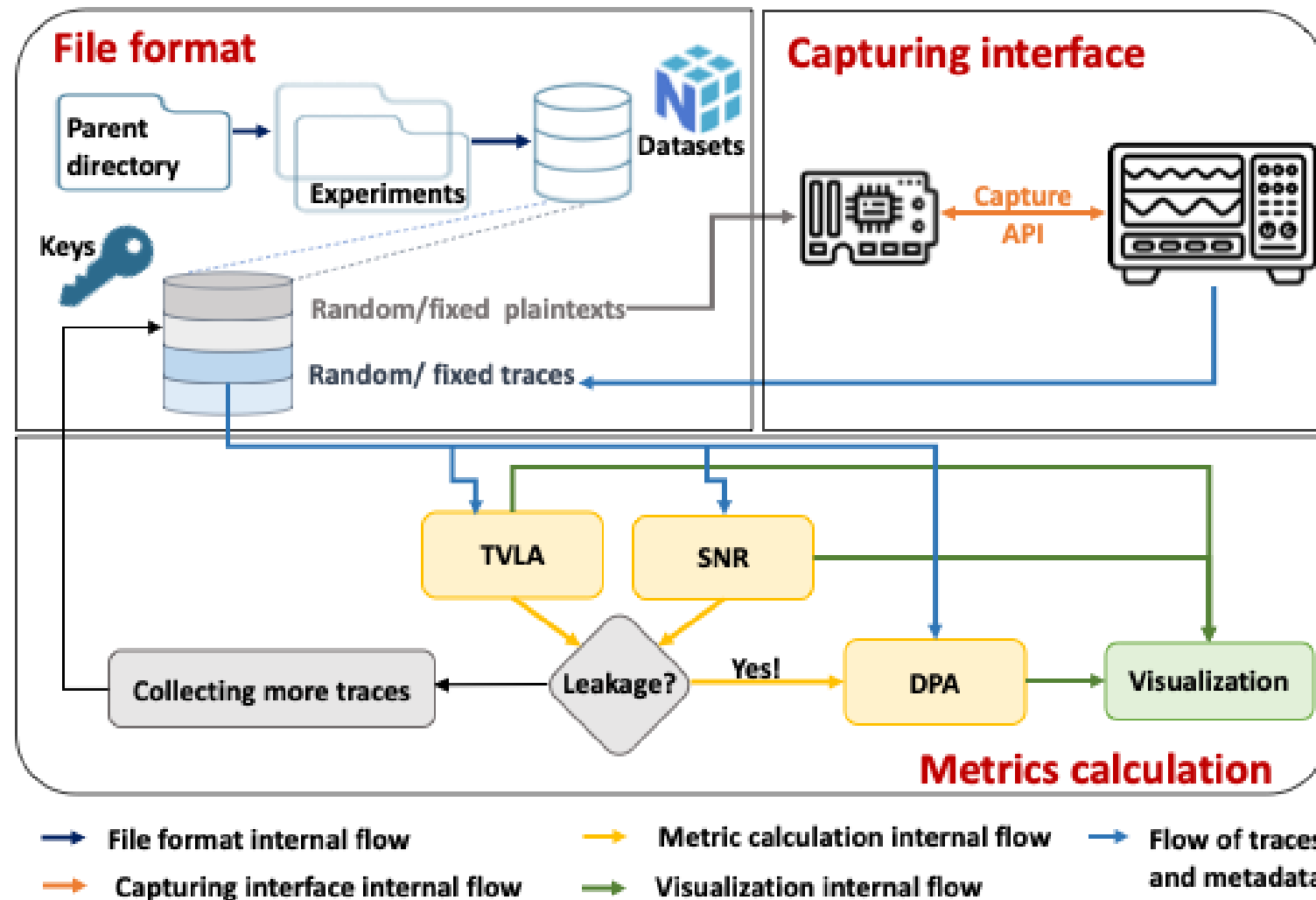
Another library?!!

- Modern targets
 - AI accelerators, PQC, and SoC
- Current challenges in SCA workflows
 - Ad hoc flows and toolchains
 - Different labs use inconsistent file formats & metric computation methods
 - Difficult to reproduce results or perform fair comparisons
 - High learning curve
 - Inaccessible toolsets and a lack of standardization hinder student adoption
 - Poor interoperability
 - Metadata is often missing or unreadable (e.g., HDF5 vs. JSON)
- Need for a unified framework
 - National security & standardization demand consistent, low-cost, reproducible SCA
 - Growing relevance in non-traditional domains (e.g., NN weights as secrets)
 - Traces are getting longer & larger (terabytes), requiring memory-aware tooling

SCAPEGoat

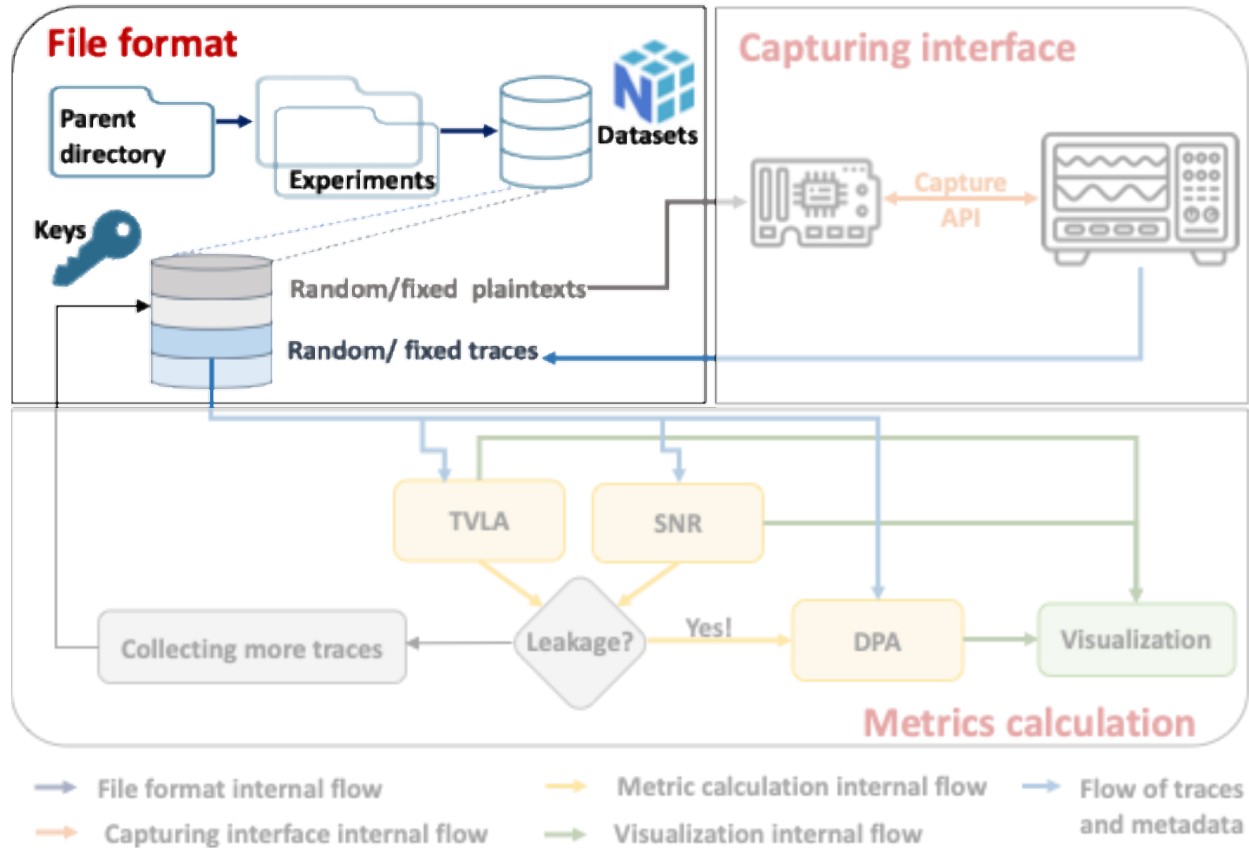
- Custom JSON-based file framework
 - Organized traces and metadata for reproducibility
 - Support for querying experiments and datasets
- Integrated capture interfaces support both high-end (Lecroy) and low-cost, e.g., Chipwhisperer tools
 - Making SCA accessible to professionals and learners
- Memory-efficient metric computation
 - Live TVLA and optimized DPA (including *2nd-order*) on partitioned datasets for scalable leakage analysis

3 modules



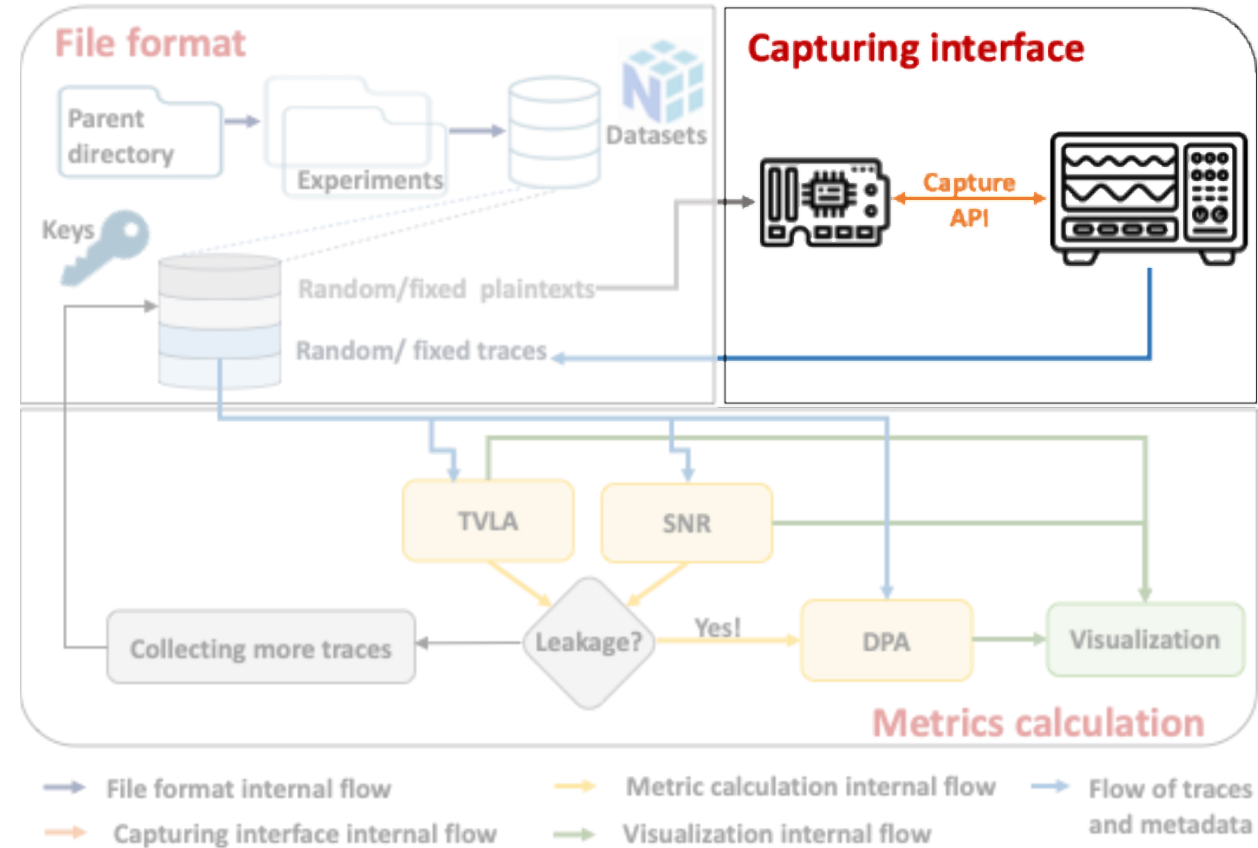
File format

- A custom file framework based on hierarchical JSON
 - Organizing experiments, datasets, and metadata in a human-readable way
- Easy adoption for existing traces
 - Flexible and compatible with other tools and workflows
- The hierarchy
 - A file parent, experiments, datasets, and a visualization folder,
 - Each layer managed through dedicated Python classes
 - Help navigate and manage large-scale SCA projects
 - supports metadata queries
- Annotation with arbitrary metadata such as environmental conditions,
- Query experiments based on these tags for reproducibility and sharing



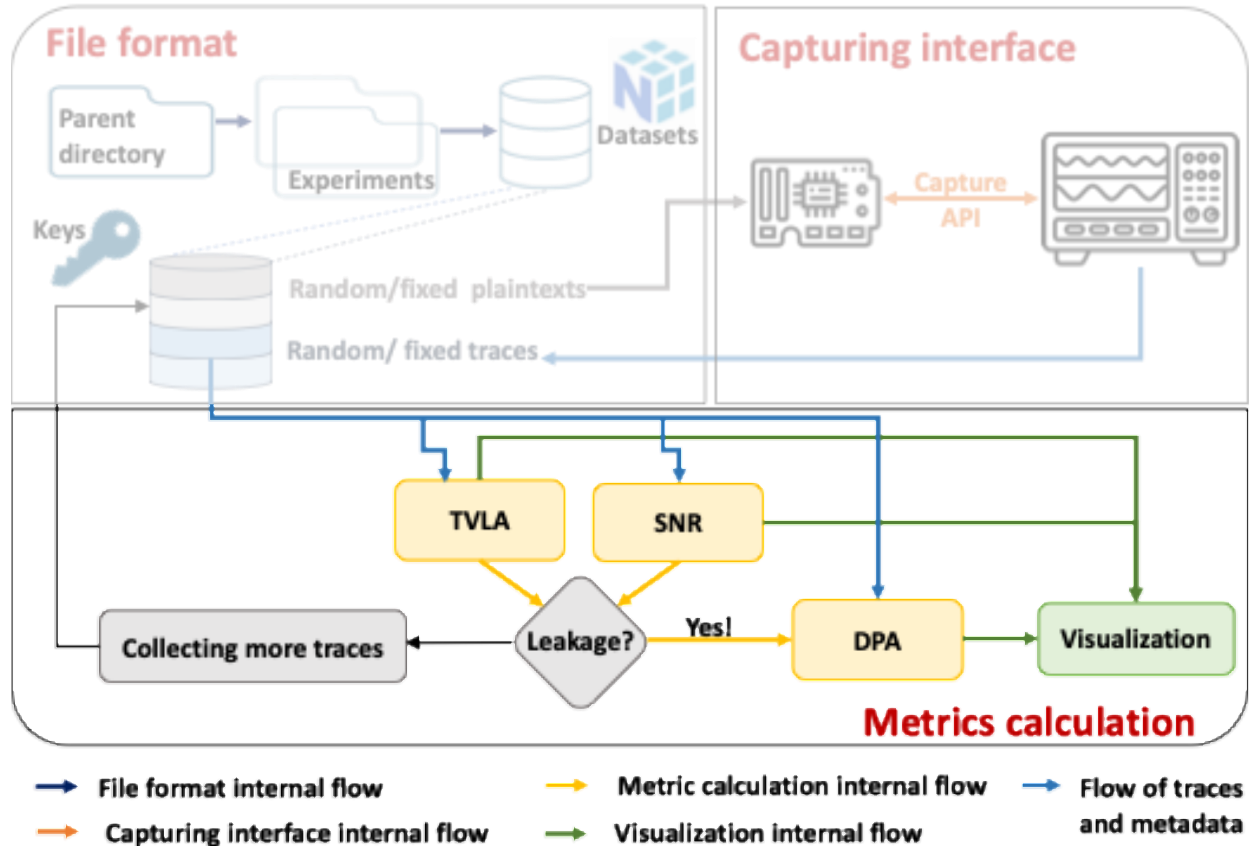
Capturing interface

- Supporting various oscilloscopes, e.g., CW and Lecroy oscilloscopes
 - PyVISA allowing integration with a wide range of commercial devices
- A dedicated, but unified, function
 - Outputs captured power traces, keys, plaintexts, and ciphertexts
 - Offering both speed and flexibility
- Automated collection and storage of fixed and random traces for TVLA



Metric computation

- Computing metrics by passing dataset names
 - Optionally store results and plots for further analysis and visualization
- Memory-efficient metric computation by processing trace partitions and accumulating results incrementally
- Live TVLA
- First-order DPA: direct leakage from individual operations
 - Effective when no countermeasures
- Second-order DPA
 - Combining multiple time leakage points to bypass protections
 - Column-wise processing in SCAPEGoat

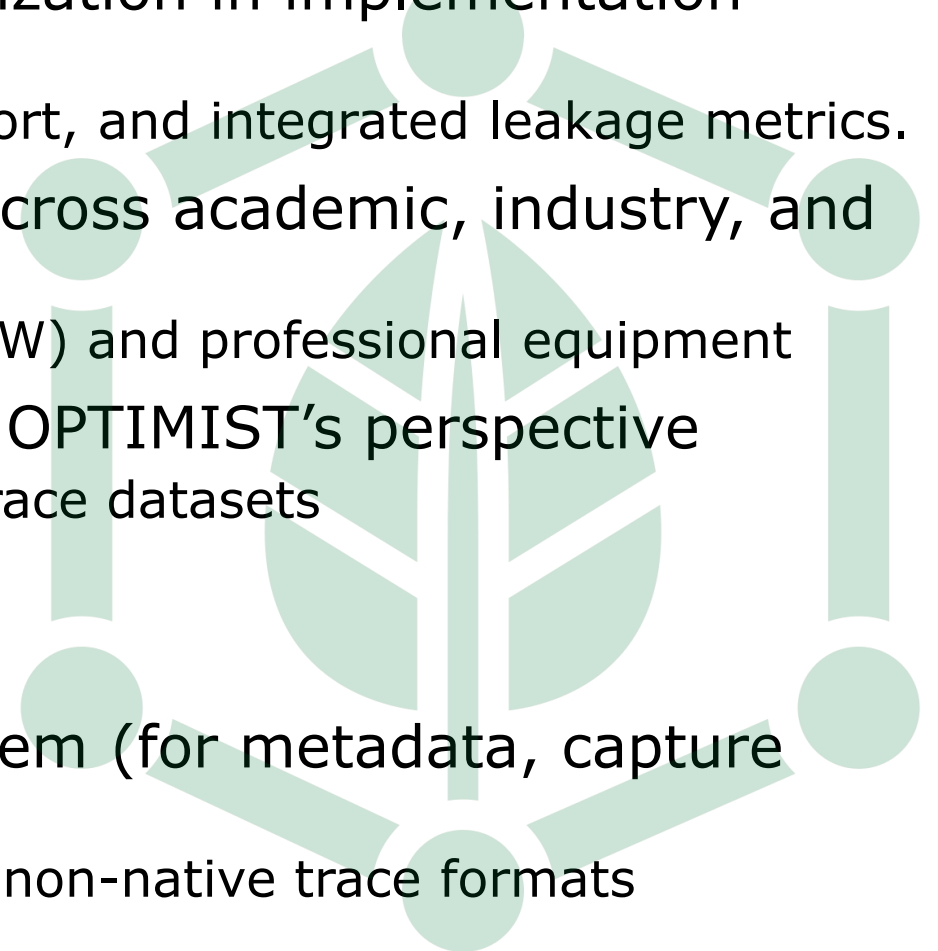


Comparison

Library	Metadata Handling	Integrated Metric Solving	Integrated File Format	Integrated Trace Capturing	Easily Expandable	DPA Support
SCAPEGoat	yes	yes	yes	yes	yes	1st/2nd order
SCALib	no	yes	no	no	yes	none
CW	no	somewhat	yes	yes	somewhat	1st order
SCARR	no	yes	yes	yes	yes	1st order
SCARED	no	yes	yes	no	no	1st order
Lascar	no	yes	yes	yes	somewhat	1st order
SEDPACK / SCAAML	yes	somewhat	yes	no	no	none

Alignment with OPTIMIST's goals

- Aiming to improve reproducibility and standardization in implementation security testing
 - Providing an open, clear file format, metadata support, and integrated leakage metrics.
- OPTIMIST's goal of accessible tools that work across academic, industry, and resource-constrained settings
 - Support for both low-cost capture hardware (e.g., CW) and professional equipment
- Scalability and reproducibility as required from OPTIMIST's perspective
 - SCAPEgoat's memory-efficient processing of large trace datasets
- Reducing barriers to entry
 - E.g., built-in implementation of standard metrics
- A component in OPTIMIST's envisioned ecosystem (for metadata, capture interfaces, trace formats etc.)
 - SCAPEGoat's modularity and support for converting non-native trace formats



Thank You

Contact Information

- Prof. Fatemeh (Saba) Ganji, Worcester Polytechnic Institute, ECE & Cybersecurity
- Webpage: <https://wp.wpi.edu/fganji/>
- E-mail: fganji@wpi.edu
- Phone: 508-831-5000